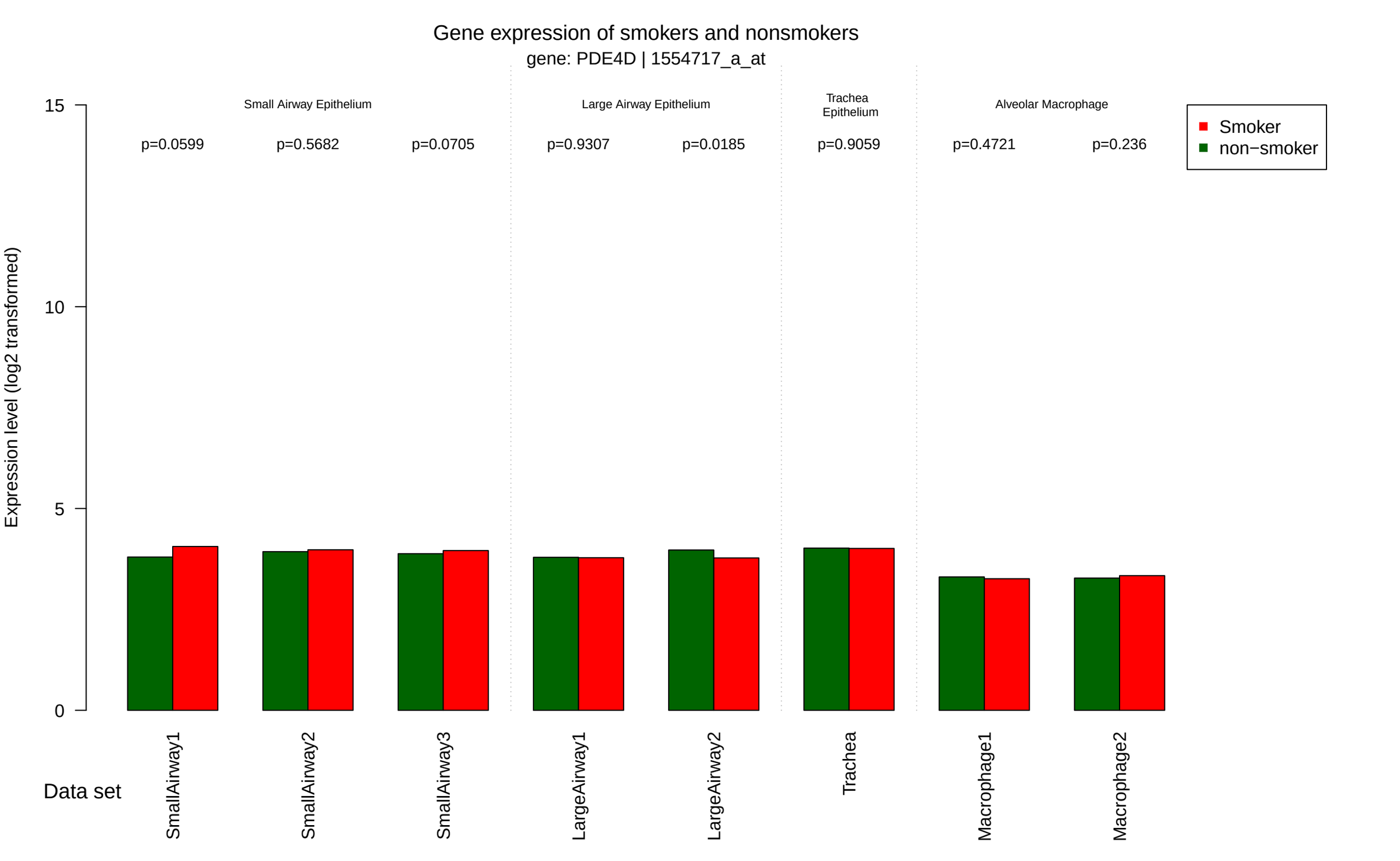
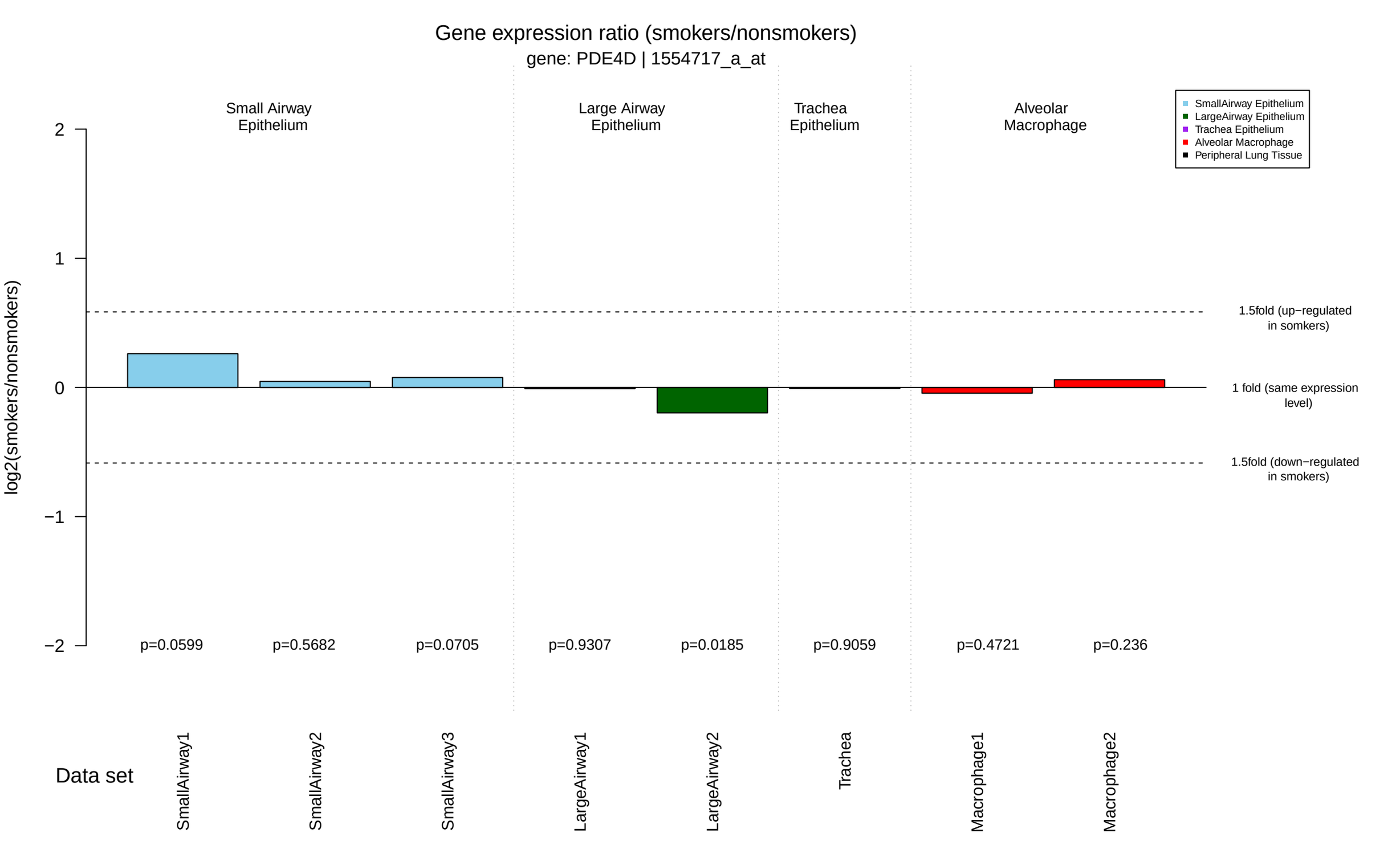
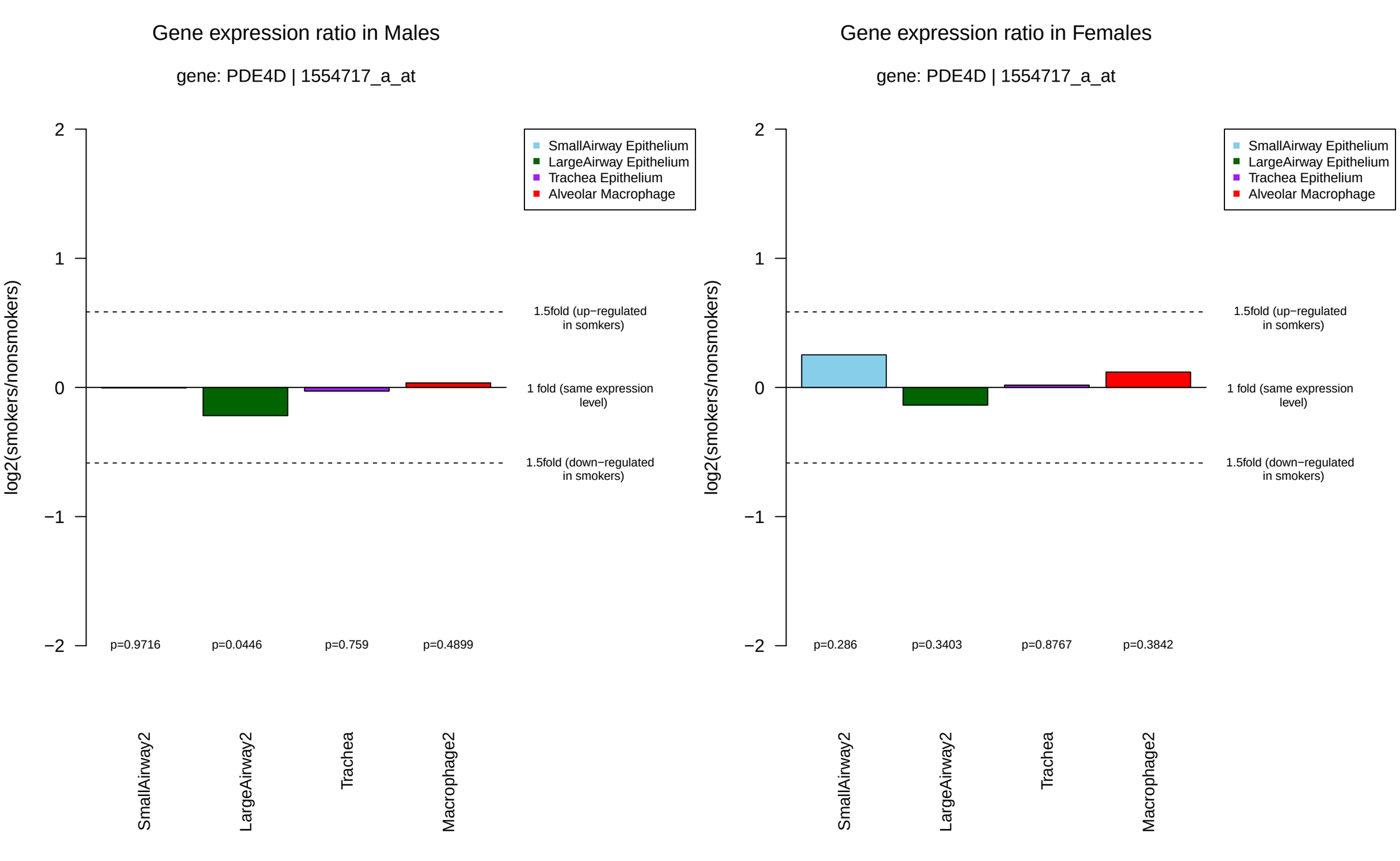
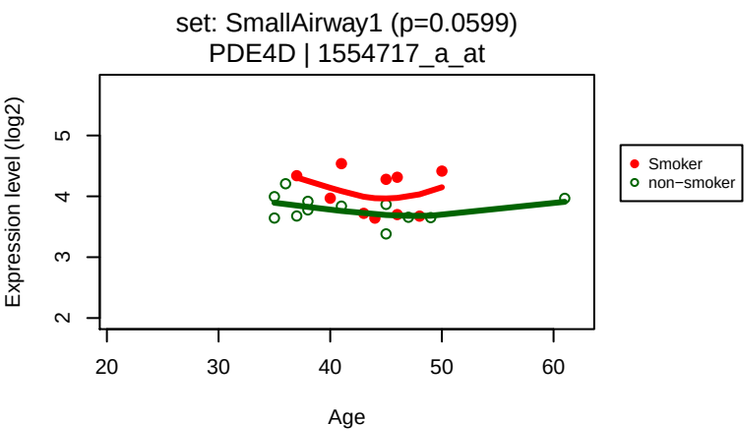
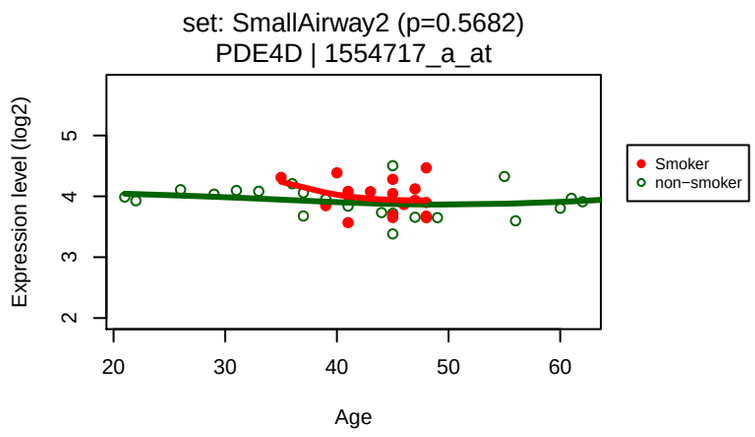
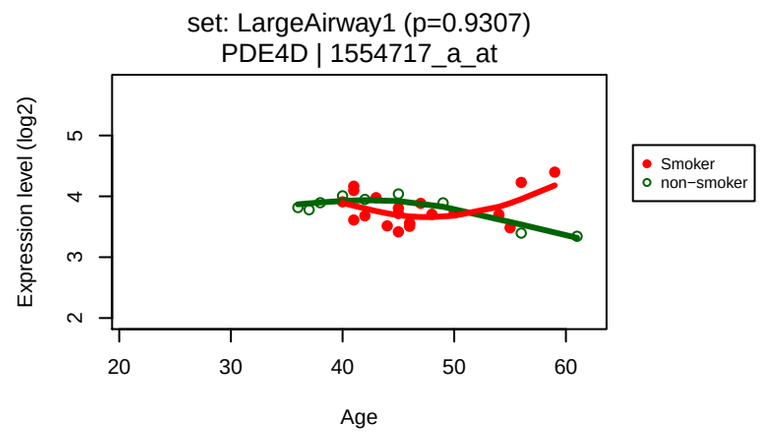

Gene expression of smokers and nonsmokers
gene: PDE4D | 1554717_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0599
p=0.5682
p=0.0705
p=0.9307
p=0.0185
p=0.9059
p=0.4721
p=0.236
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

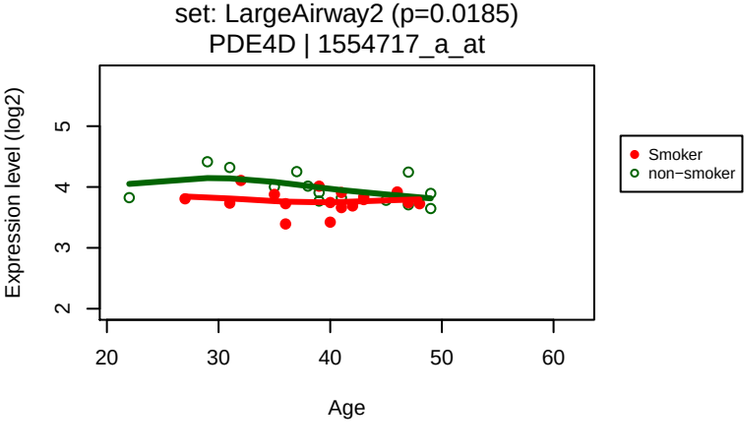

Gene expression ratio (smokers/nonsmokers)
gene: PDE4D | 1554717_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0599
p=0.5682
p=0.0705
p=0.9307
p=0.0185
p=0.9059
p=0.4721
p=0.236
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

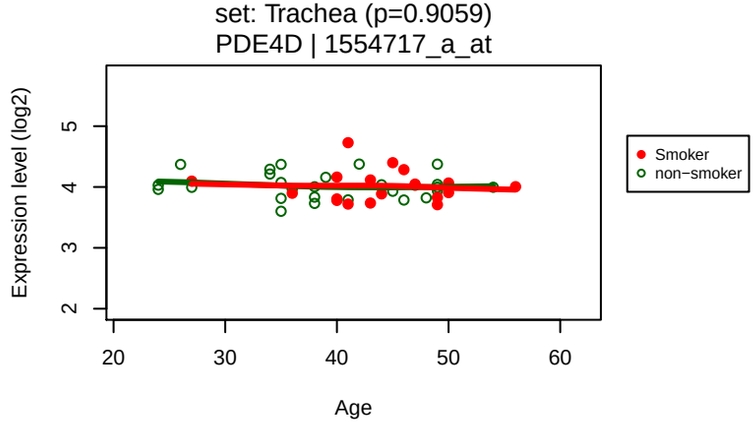

Gene expression ratio in Males
gene: PDE4D | 1554717_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9716
p=0.0446
p=0.759
p=0.4899
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE4D | 1554717_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.286
p=0.3403
p=0.8767
p=0.3842
SmallAirway2
LargeAirway2
Trachea
Macrophage2

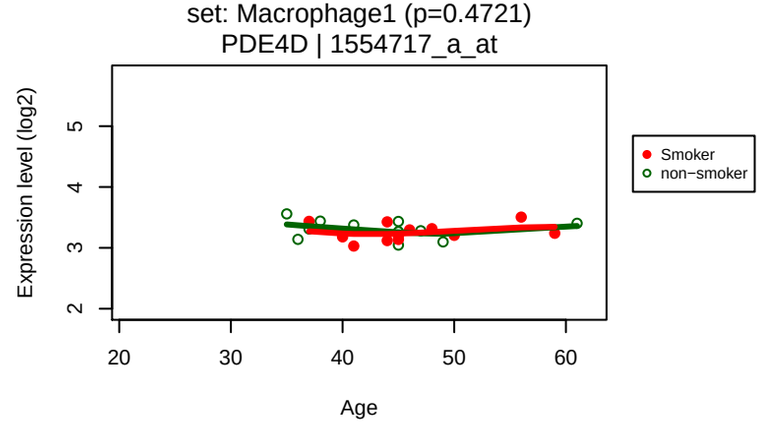

set: SmallAirway1 (p=0.0599)
PDE4D | 1554717_a_at
Expression level (log2)
Age
Smoker
non−smoker

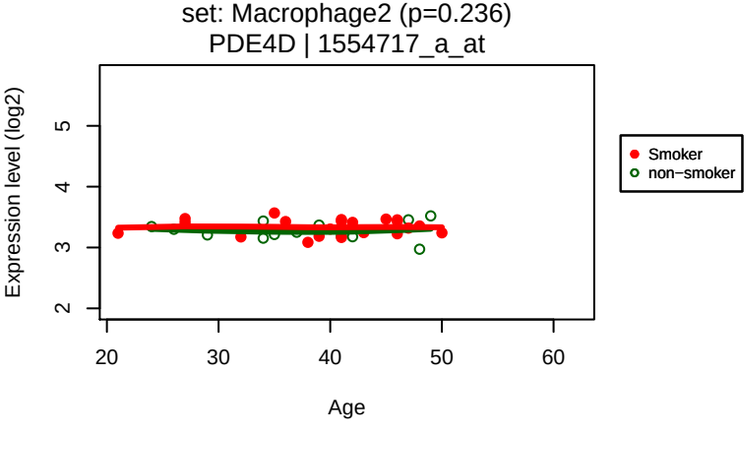

set: SmallAirway2 (p=0.5682)
PDE4D | 1554717_a_at
Expression level (log2)
Age
Smoker
non−smoker

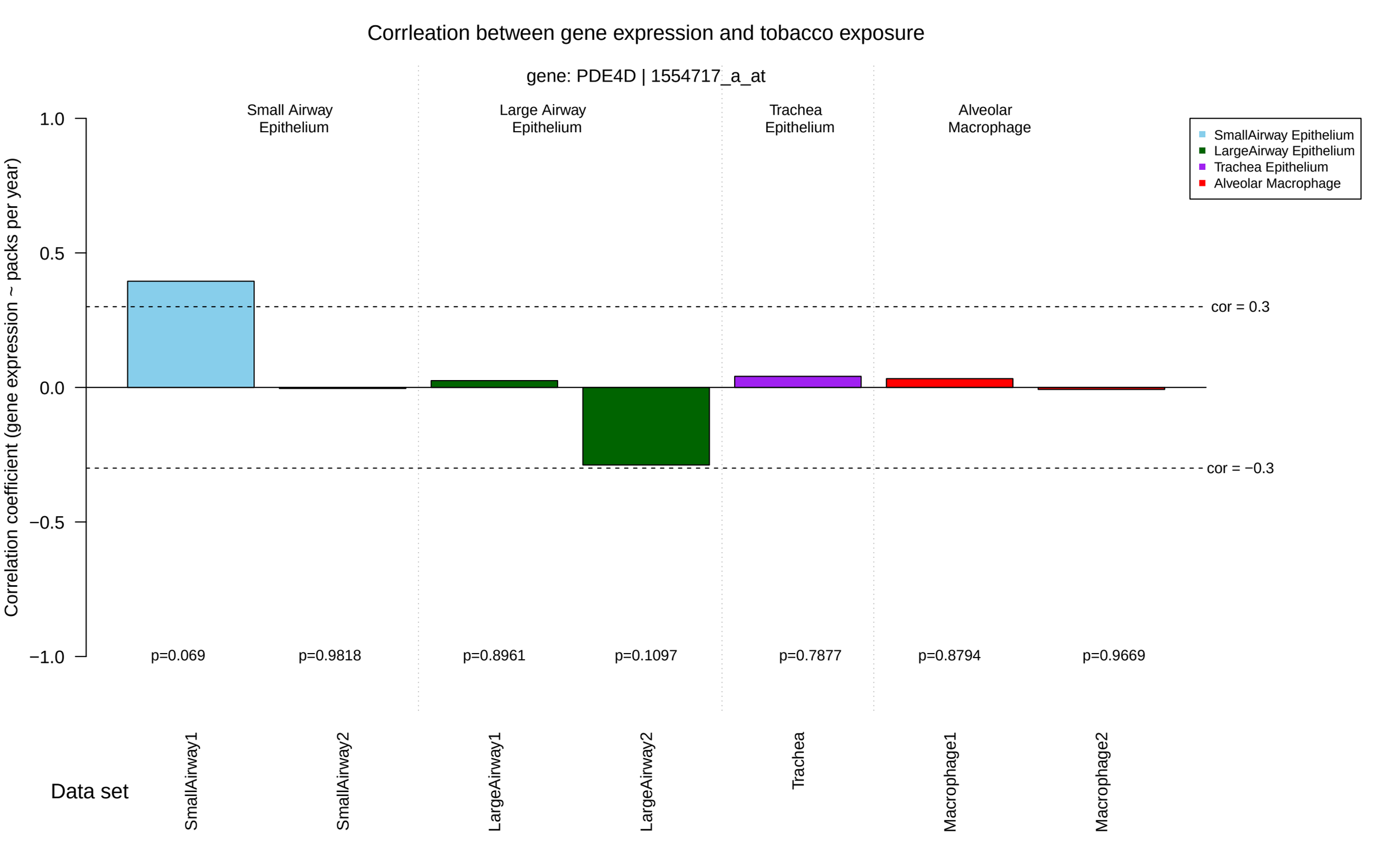

set: LargeAirway1 (p=0.9307)
PDE4D | 1554717_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0185)
PDE4D | 1554717_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9059)
PDE4D | 1554717_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4721)
PDE4D | 1554717_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.236)
PDE4D | 1554717_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE4D | 1554717_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.069
p=0.9818
p=0.8961
p=0.1097
p=0.7877
p=0.8794
p=0.9669
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2